Aptitude

1️⃣ Time, Speed & Distance

Formulas

1. Speed $= \frac{\text{Distance}}{\text{Time}}$
2. Time $= \frac{\text{Distance}}{\text{Speed}}$
3. Distance $=$ Speed $\times$ Time
4. Relative speed (same direction) = $S_1 - S_2$
5. Relative speed (opposite direction) = $S_1 + S_2$

Short Tricks

- Convert **km/hr → m/s**: multiply by $\frac{5}{18}$
- Convert **m/s → km/hr**: multiply by $\frac{18}{5}$
- Train crossing a pole: Time $= \frac{\text{Length of train}}{\text{Speed}}$
- Two objects moving towards each other: use **sum of speeds**
- Two objects moving in same direction: use **difference of speeds**

2️⃣ Time & Work / Pipes and Cisterns

Formulas

1. Work done = Rate × Time
2. If A can do a work in $a$days, daily work = $\frac{1}{a}$
3. Work together:

$$\text{Days} = \frac{1}{\text{Sum of daily work rates}}$$

4. Pipes filling:
   - Fill rate = $\frac{\text{Capacity}}{\text{Time to fill}}$
   - Net rate if both fill & drain = Fill rate – Drain rate

Short Tricks

- If A finishes work in 10 days & B in 15 → together: $1/10 + 1/15 = 1/6$→ 6 days
- For efficiency problems: Work $\propto$ Number of workers $\times$ Time

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3️⃣ Percentages / Profit & Loss / Discount

Formulas

1. Percentage = $\frac{\text{Part}}{\text{Whole}} \times 100$
2. Profit = Selling Price – Cost Price
3. Loss = Cost Price – Selling Price
4. Profit% = $\frac{\text{Profit}}{\text{Cost Price}} \times 100$
5. Loss% = $\frac{\text{Loss}}{\text{Cost Price}} \times 100$
6. Discount% = $\frac{\text{Discount}}{\text{Marked Price}} \times 100$

Short Tricks

- Quick profit/loss trick:
  - CP = 100 → Profit 20% → SP = 120
  - CP = 100 → Loss 10% → SP = 90
- Successive discounts: multiply factors
  - e.g., 20% & 10% → $0.8 \times 0.9 = 0.72$→ 28% total discount

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4️⃣ Ratio & Proportion / Averages / Simple & Compound Interest

Formulas

- **Ratio:** $a:b = \frac{a}{b}$
- **Proportion:** $a:b = c:d \Rightarrow ad = bc$
- **Average:** Average $= \frac{\text{Sum of terms}}{\text{Number of terms}}$
- **Simple Interest:** $SI = \frac{P \cdot R \cdot T}{100}$
- **Compound Interest:** $CI = P\left(1+\frac{R}{100}\right)^{T} - P$

Short Tricks

- Weighted average: multiply each quantity by weight, sum, divide by total weight
- CI ≈ SI + (SI × R)/100 (quick approximation for 2 years)
- Ratio sharing: divide total in ratio parts → Part $= \frac{\text{Total}\times\text{Ratio part}}{\text{Sum of ratios}}$

5️⃣ Number Series / Series Completion (Logical Reasoning)

Types of Series

1. **Arithmetic:** difference constant → add/subtract
2. **Geometric:** ratio constant → multiply/divide
3. **Square/Cube series:** n^2, n^3 patterns
4. **Mixed series:** alternate arithmetic/geometric

Short Tricks

- Check **difference of differences** → helps in 2nd order arithmetic series
- Look for **pattern in operations**: +, –, ×, ÷, powers, factorial
- Check **alternate series** → odd/even index separately

CS Fundamentals

Topics Covered

1. **Data Structures –** Arrays, Strings, Linked Lists, Stacks, Queues, Trees
2. **Algorithms & Complexity** – Sorting, Searching, Time & Space Complexity
3. **Programming Basics –** Loops, Functions, Recursion, Output Prediction
4. **OOP Concepts –** Classes, Objects, Inheritance, Polymorphism, Encapsulation, Abstraction
5. **DBMS & SQL** – Keys, Normalization, Queries, Transactions
6. **Operating Systems** – CPU Scheduling, Threads, Memory, Deadlocks
7. **Computer Networks** – OSI/TCP-IP, Protocols, IP Addresses
8. **Web/REST Basics –** HTTP Methods, REST principles

Sample Format for Each MCQ

Q1. What is the time complexity of binary search in a sorted array of size n?
A) O(n)
B) O(log n)
C) O(n log n)
D) O(1)

Answer: B) O(log n)

Explanation: Binary search divides the search space by 2 in each step, so time complexity is logarithmic.

I can generate the **full 100 technical MCQs** in a **single downloadable document** or as a **series of messages here** with answers + explanations.

Since 100 questions in one message may be too long for the chat, the **best way** is to split them into **batches of 20–25 MCQs**.

Do you want me to **start with the first 25 MCQs** now?

yes

Perfect! Here's the **first batch of 25 technical MCQs** for ConnectWise-style internship tests, with **options, correct answers, and brief explanations**.

Batch 1: Technical MCQs (1–25)

Q1. What is the time complexity of binary search in a sorted array of size n?
A) O(n)
B) O(log n)
C) O(n log n)
D) O(1)

Answer: B) O(log n)
Explanation: Binary search halves the search space each time → logarithmic time.

Q2. Which data structure uses **FIFO (First In First Out)** principle?
A) Stack
B) Queue
C) Linked List
D) Tree

Answer: B) Queue
Explanation: Queue inserts at rear and deletes from front (FIFO).

Q3. What is the output of the following C code snippet?

```
int x = 5;

printf("%d", x++);
```

A) 4
B) 5
C) 6
D) Error

Answer: B) 5
Explanation: Post-increment returns the current value first, then increments.

Q4. Which sorting algorithm is the fastest on average for large datasets?
A) Bubble Sort
B) Insertion Sort
C) Merge Sort
D) Selection Sort

Answer: C) Merge Sort
Explanation: Merge Sort has O(n log n) average and worst-case complexity.

Q5. In OOP, which concept allows a class to **take many forms**?
A) Inheritance
B) Polymorphism
C) Encapsulation
D) Abstraction

Answer: B) Polymorphism
Explanation: Polymorphism allows a method to behave differently based on object type.

Q6. Which of the following is **not a primary key property** in DBMS?
A) Unique
B) Not Null
C) Can be duplicated
D) Identifies a record

Answer: C) Can be duplicated
Explanation: Primary keys must uniquely identify each row.

Q7. Which traversal method is used to **print a binary tree in level order**?
A) Preorder
B) Inorder
C) Postorder
D) Breadth-First

Answer: D) Breadth-First
Explanation: Level-order traversal visits nodes level by level.

Q8. Which SQL command is used to remove duplicates from a query result?
A) UNIQUE
B) DISTINCT
C) REMOVE
D) DELETE

Answer: B) DISTINCT
Explanation: DISTINCT eliminates duplicate rows in SELECT output.

Q9. Which OS scheduling algorithm is **non-preemptive**?
A) Round Robin
B) Shortest Job First (SJF)
C) First Come First Serve (FCFS)
D) Priority (preemptive)

Answer: C) First Come First Serve (FCFS)
Explanation: FCFS executes processes in arrival order without preemption.

Q10. What is the output of this Python code?

```
x = [1,2,3]

print(x*2)
```

A) [1,2,3,1,2,3]
B) [2,4,6]
C) [1,2,3,2]
D) Error

Answer: A) [1,2,3,1,2,3]
Explanation: Multiplying a list in Python repeats its elements.

Q11. Which data structure is used for **function call management** in recursion?
A) Queue
B) Stack
C) Linked List
D) Graph

Answer: B) Stack
Explanation: The call stack stores function calls in LIFO order.

Q12. Which HTTP method is used to **update a resource completely**?
A) GET
B) POST
C) PUT
D) DELETE

Answer: C) PUT
Explanation: PUT replaces an existing resource with new data.

Q13. In a min-heap, the **minimum element** is:
A) At the root
B) At the leaf
C) In the middle
D) Can be anywhere

Answer: A) At the root
Explanation: By definition, min-heap root stores the smallest value.

Q14. Which of the following is **not a feature of OOP**?
A) Inheritance
B) Polymorphism
C) Recursion
D) Abstraction

Answer: C) Recursion
Explanation: Recursion is a programming technique, not an OOP concept.

Q15. What is the best case complexity of Quick Sort?
A) $O(n^2)$
B) O(n log n)
C) O(log n)
D) O(n)

Answer: B) O(n log n)
Explanation: Best case occurs when the pivot divides array evenly.

Q16. Which SQL normal form removes **partial dependency**?
A) 1NF
B) 2NF
C) 3NF
D) BCNF

Answer: B) 2NF
Explanation: 2NF ensures all non-key attributes are fully dependent on the primary key.

Q17. Which process state means a process is **waiting for an event**?
A) Ready
B) Running
C) Blocked
D) New

Answer: C) Blocked
Explanation: A blocked process waits for I/O or a resource.

Q18. Which layer in OSI model is responsible for **end-to-end delivery**?
A) Transport
B) Network
C) Data Link
D) Physical

Answer: A) Transport
Explanation: Transport layer ensures complete data delivery using TCP/UDP.

Q19. What is the output of the following Java code?

int a = 5;

System.out.println(++a);

A) 4
B) 5
C) 6
D) Error

Answer: C) 6
Explanation: Pre-increment increments the value before printing.

Q20. In a queue implemented using **array**, if front = 0, rear = n-1, and a new element is inserted, what happens?
A) Overflow
B) Underflow
C) Successful insertion
D) Depends on programming language

Answer: A) Overflow
Explanation: In a fixed-size array queue, rear at n-1 means queue is full.

Q21. Which of the following is a **non-linear data structure**?
A) Array
B) Linked List
C) Tree
D) Queue

Answer: C) Tree
Explanation: Trees have hierarchical structure; others are linear.

Q22. Which network device is used to **connect two different networks**?
A) Hub
B) Switch
C) Router
D) Repeater

Answer: C) Router
Explanation: Routers forward packets between networks.

Q23. Which of the following is a **characteristic of REST API**?
A) Stateful
B) Stateless
C) Requires server session
D) Uses SOAP

Answer: B) Stateless
Explanation: REST APIs do not store client session state on the server.

Q24. Which data structure is used in **Breadth-First Search (BFS)** of a graph?
A) Stack
B) Queue
C) Linked List
D) Hash Map

Answer: B) Queue
Explanation: BFS uses a queue to explore nodes level by level.

Q25. What is the **space complexity** of Merge Sort?
A) O(1)
B) O(log n)
C) O(n)
D) O(n log n)

Answer: C) O(n)
Explanation: Merge Sort requires extra array of size n for merging.

Batch 2: Technical MCQs (26–50)

Q26. Which of the following is **not a primitive data type in Java**?
A) int
B) float
C) String
D) boolean

Answer: C) String
Explanation: String is a class in Java, not a primitive type.

Q27. In a max-heap, where is the **largest element** located?
A) Root
B) Leaf
C) Middle
D) Random

Answer: A) Root
Explanation: Max-heap stores the largest value at the root by definition.

Q28. Which of the following is **LIFO** in nature?
A) Queue
B) Stack
C) Linked List
D) Graph

Answer: B) Stack
Explanation: Stack inserts and deletes from the same end (Last In First Out).

Q29. What is the **default access modifier** for members of a class in Java?
A) private
B) public
C) protected
D) package-private

Answer: D) package-private
Explanation: Members without any modifier are accessible within the same package.

Q30. Which of the following algorithms is **in-place**?
A) Merge Sort
B) Quick Sort
C) Heap Sort
D) Both B and C

Answer: D) Both B and C
Explanation: Quick Sort and Heap Sort sort using constant or O(log n) extra space; Merge Sort uses O(n).

Q31. In Python, which of the following is a **mutable data type**?
A) Tuple
B) List
C) String
D) Integer

Answer: B) List
Explanation: Lists can be modified after creation; tuples and strings are immutable.

Q32. What does SQL **JOIN** do?
A) Combines rows from two or more tables
B) Deletes rows

C) Updates rows
D) Creates a new table

Answer: A) Combines rows from two or more tables
Explanation: JOIN combines data based on a related column between tables.

Q33. Which process state occurs **just before execution starts**?
A) Ready
B) Running
C) Waiting
D) New

Answer: D) New
Explanation: Process moves from New → Ready → Running.

Q34. What is the output of this C++ code?

```
int a = 10;

int b = 3;

cout << a % b;
```

A) 3
B) 1
C) 0
D) 10

Answer: B) 1
Explanation: % is modulo operator → remainder of 10 ÷ 3 is 1.

Q35. Which of the following is **not true about stack**?
A) Uses LIFO principle
B) Insertion and deletion at one end
C) Can access elements randomly
D) Used for function calls

Answer: C) Can access elements randomly
Explanation: Stack allows only top element access; no random access.

Q36. Which OS scheduling algorithm is **preemptive and uses time slice**?
A) FCFS
B) SJF
C) Round Robin
D) Priority Non-Preemptive

Answer: C) Round Robin
Explanation: Round Robin allocates CPU for fixed time slices (quantum).

Q37. Which of the following is a **non-linear search** algorithm?
A) Linear Search
B) Binary Search
C) Depth-First Search
D) Selection Sort

Answer: C) Depth-First Search
Explanation: DFS is used for graphs/trees (non-linear structures).

Q38. Which SQL command is used to **remove all rows from a table but keep its structure**?
A) DELETE
B) DROP
C) TRUNCATE
D) REMOVE

Answer: C) TRUNCATE
Explanation: TRUNCATE clears all data but keeps table schema.

Q39. What is the output of the following Python snippet?

```
a = [1,2,3]

b = a

b.append(4)

print(a)
```

A) [1,2,3]
B) [1,2,3,4]
C) [4]
D) Error

Answer: B) [1,2,3,4]
Explanation: b references the same list as a; changes reflect in both.

Q40. In OOP, **which concept hides internal implementation details?**
A) Polymorphism
B) Inheritance
C) Encapsulation
D) Abstraction

Answer: C) Encapsulation
Explanation: Encapsulation wraps data and methods, controlling access.

Q41. Which is **non-preemptive scheduling**?
A) SJF (preemptive)
B) FCFS
C) Round Robin
D) Priority Preemptive

Answer: B) FCFS
Explanation: FCFS executes tasks in arrival order without interruption.

Q42. Which layer in TCP/IP is responsible for **routing packets**?
A) Application
B) Transport
C) Internet
D) Physical

Answer: C) Internet
Explanation: Internet layer handles addressing and routing (IP layer).

Q43. What is the output of this JavaScript code?

console.log(typeof null);

A) null
B) object
C) undefined
D) number

Answer: B) object
Explanation: In JavaScript, typeof null returns object (legacy quirk).

Q44. Which of the following sorting algorithms is **stable**?
A) Quick Sort
B) Heap Sort
C) Merge Sort
D) Selection Sort

Answer: C) Merge Sort
Explanation: Merge Sort preserves relative order of equal elements.

Q45. Which of the following is **true about linked lists**?
A) Fixed size in memory
B) Contiguous memory allocation
C) Dynamic size, elements connected via pointers
D) Random access possible

Answer: C) Dynamic size, elements connected via pointers
Explanation: Linked lists use nodes and pointers, size is dynamic.

Q46. Which of these is a **synchronous protocol**?
A) UDP
B) TCP
C) HTTP
D) FTP

Answer: B) TCP
Explanation: TCP is connection-oriented and synchronous (reliable delivery).

Q47. Which of the following is **immutable in Java**?
A) String
B) StringBuilder
C) ArrayList
D) HashMap

Answer: A) String
Explanation: Java strings cannot be modified after creation.

Q48. What is the **time complexity of linear search** in worst case?
A) O(1)
B) O(n)
C) O(log n)
D) $O(n^2)$

Answer: B) O(n)
Explanation: Linear search may need to check all n elements.

Q49. Which of the following **TCP/IP layers is responsible for end-to-end delivery**?
A) Network
B) Transport
C) Data Link
D) Physical

Answer: B) Transport
Explanation: Transport layer (TCP/UDP) ensures complete data delivery.

Q50. Which of the following is **NOT a feature of REST API**?
A) Stateless
B) Client-server architecture
C) Supports caching
D) Requires server session

Answer: D) Requires server session
Explanation: REST APIs are stateless and do not store server-side sessions.

Batch 3: Technical MCQs (51–75)

Q51. Which of the following is **true about a stack implemented using an array**?
A) Stack size is dynamic
B) Stack size is fixed
C) Stack cannot overflow
D) Stack supports random access

Answer: B) Stack size is fixed
Explanation: Array-based stacks have a fixed maximum size; dynamic stacks use linked lists.

Q52. Which of the following is a **preemptive CPU scheduling algorithm**?
A) FCFS
B) SJF (Shortest Job First)
C) Round Robin
D) FCFS Non-preemptive

Answer: C) Round Robin
Explanation: Round Robin can preempt the running process when the time quantum expires.

Q53. Which of the following is **true for a binary search tree (BST)?**
A) Left child > root, right child < root
B) Left child < root, right child > root
C) Left child = root, right child = root
D) Nodes can be in any order

Answer: B) Left child < root, right child > root
Explanation: BST property: left subtree < root < right subtree.

Q54. What is the **output of this Python snippet?**

```
x = {1, 2, 3}

x.add(2)

print(x)
```

A) {1, 2, 3}
B) {1, 2, 2, 3}
C) {2, 3}
D) Error

Answer: A) {1, 2, 3}
Explanation: Sets store unique elements; adding 2 again has no effect.

Q55. Which of the following **cannot be inherited in Java**?
A) Private methods
B) Public methods
C) Protected methods
D) Default methods in same package

Answer: A) Private methods
Explanation: Private members are not visible to subclasses.

Q56. Which of the following **is a linear data structure?**
A) Stack
B) Tree
C) Graph
D) Heap

Answer: A) Stack
Explanation: Stack elements are arranged linearly (LIFO).

Q57. Which of the following **is a non-linear data structure?**
A) Array
B) Linked List
C) Tree
D) Queue

Answer: C) Tree
Explanation: Tree has hierarchical structure; other options are linear.

Q58. What is the output of the following Java code?

```
int x = 10;

System.out.println(x--);
```

A) 9
B) 10
C) 11
D) Error

Answer: B) 10
Explanation: Post-decrement prints current value first, then decrements.

Q59. Which of the following is **true about recursion**?
A) Requires iterative loop
B) Always slower than loops
C) Function calls itself
D) Cannot return values

Answer: C) Function calls itself
Explanation: Recursion is a function calling itself to solve subproblems.

Q60. Which of the following is the **main advantage of linked lists over arrays?**
A) Random access
B) Dynamic size
C) Contiguous memory
D) Faster search

Answer: B) Dynamic size
Explanation: Linked lists can grow/shrink at runtime; arrays have fixed size.

Q61. Which of the following is **not a component of CPU?**
A) ALU
B) CU
C) RAM
D) Registers

Answer: C) RAM
Explanation: RAM is memory; CPU contains ALU, CU, and registers.

Q62. Which of the following **TCP/IP layers maps to the OSI Transport layer?**
A) Application
B) Internet
C) Transport
D) Network Access

Answer: C) Transport
Explanation: TCP/UDP operate at Transport layer providing end-to-end delivery.

Q63. Which of the following **sorting algorithms is not in-place?**
A) Quick Sort
B) Heap Sort
C) Merge Sort
D) Bubble Sort

Answer: C) Merge Sort
Explanation: Merge Sort requires extra O(n) memory for merging arrays.

Q64. What is the **default value of a boolean variable** in Java?
A) true
B) false
C) 0
D) null

Answer: B) false
Explanation: Boolean defaults to false in Java if not initialized.

Q65. Which of the following is **true about TCP**?
A) Connectionless
B) Provides reliable delivery
C) Unreliable
D) Stateless

Answer: B) Provides reliable delivery
Explanation: TCP ensures data is delivered reliably with acknowledgment.

Q66. What is the output of the following Python snippet?

print(2 ** 3 ** 2)

A) 64
B) 512
C) 256
D) Error

Answer: B) 512
Explanation: Exponentiation is right-associative → 3**2 = 9 → 2**9 = 512.

Q67. Which of the following is a **non-preemptive scheduling algorithm**?
A) FCFS
B) Round Robin
C) Priority Preemptive
D) SRTF

Answer: A) FCFS
Explanation: First Come First Serve executes tasks until completion without interruption.

Q68. Which SQL clause is used to **sort query results**?
A) ORDER BY
B) GROUP BY
C) SORT
D) FILTER

Answer: A) ORDER BY
Explanation: ORDER BY sorts result set by specified columns.

Q69. Which of the following **is not a feature of OOP**?
A) Encapsulation
B) Polymorphism
C) Abstraction
D) Recursion

Answer: D) Recursion
Explanation: Recursion is a programming technique, not an OOP feature.

Q70. Which Python collection **maintains insertion order and allows duplicates**?
A) Set
B) List
C) Dictionary
D) Tuple

Answer: B) List
Explanation: Lists maintain order and allow repeated elements.

Q71. What is the **output of the following C++ snippet?**

```
int a = 5;

int b = 2;

cout << a / b;
```

A) 2.5
B) 2
C) 3
D) Error

Answer: B) 2
Explanation: Integer division truncates decimal → 5 / 2 = 2.

Q72. Which data structure is used in **Depth-First Search (DFS)**?
A) Stack
B) Queue
C) Priority Queue
D) Hash Table

Answer: A) Stack
Explanation: DFS uses LIFO behavior; can be implemented via recursion (implicit stack) or explicit stack.

Q73. Which of the following is **true about Merge Sort**?
A) Stable, O(n log n), not in-place
B) Unstable, O(n log n), in-place
C) Stable, $O(n^2)$, in-place
D) Unstable, $O(n^2)$, not in-place

Answer: A) Stable, O(n log n), not in-place
Explanation: Merge Sort preserves order, takes O(n log n), and requires extra memory.

Q74. Which OSI layer is responsible for **error detection and correction**?
A) Data Link
B) Network
C) Transport
D) Physical

Answer: A) Data Link
Explanation: Data Link layer ensures correct frame transmission using CRC/checksum.

Q75. Which of the following is **mutable in Python**?
A) Tuple
B) String
C) List
D) Int

Answer: C) List
Explanation: Lists can be changed after creation; tuples, strings, and ints are immutable.

Batch 4: Technical MCQs (76–100)

Q76. Which of the following is a **stable sorting algorithm**?
A) Quick Sort
B) Merge Sort
C) Heap Sort
D) Selection Sort

Answer: B) Merge Sort
Explanation: Merge Sort maintains the relative order of equal elements.

Q77. In a **circular queue**, how do we determine that the queue is full?
A) rear = front
B) rear + 1 = front

C) front = -1
D) rear = -1

Answer: B) rear + 1 = front
Explanation: Circular queue is full when incrementing rear wraps around to front.

Q78. Which of the following is **true for a doubly linked list**?
A) Can traverse only forward
B) Can traverse forward and backward
C) Cannot delete nodes
D) Elements are stored in contiguous memory

Answer: B) Can traverse forward and backward
Explanation: Doubly linked list has next and previous pointers.

Q79. Which of the following **is not a hashing collision resolution technique**?
A) Chaining
B) Open addressing
C) Linear probing
D) Quick Sort

Answer: D) Quick Sort
Explanation: Quick Sort is a sorting algorithm, not a collision resolution technique.

Q80. Which layer of TCP/IP **directly interacts with the application**?
A) Transport
B) Network
C) Application
D) Internet

Answer: C) Application
Explanation: Application layer provides interface for user applications (HTTP, FTP).

Q81. Which of the following **is a non-preemptive scheduling algorithm**?
A) Round Robin
B) Priority (Preemptive)
C) FCFS
D) Shortest Remaining Time First

Answer: C) FCFS
Explanation: FCFS executes processes in arrival order until completion.

Q82. In SQL, which of the following **removes a table completely including its structure**?
A) DELETE

B) TRUNCATE
C) DROP
D) REMOVE

Answer: C) DROP
Explanation: DROP deletes table and schema permanently.

Q83. Which of the following is **true about polymorphism** in OOP?
A) Overloading and overriding are examples
B) Hides internal details
C) Only applicable in C
D) Increases memory usage

Answer: A) Overloading and overriding are examples
Explanation: Polymorphism allows objects to take multiple forms via overloading/overriding.

Q84. Which of the following **is not a characteristic of a set in Python**?
A) Unordered
B) Mutable
C) Allows duplicates
D) Iterable

Answer: C) Allows duplicates
Explanation: Sets store unique elements only.

Q85. Which data structure is used in **implementation of recursion**?
A) Queue
B) Stack
C) Linked List
D) Graph

Answer: B) Stack
Explanation: Recursive calls are pushed onto the call stack.

Q86. Which of the following is **true about heap sort**?
A) O(n log n), in-place, unstable
B) O(n^2), stable, in-place
C) O(n log n), not in-place, stable
D) O(n^2), unstable, not in-place

Answer: A) O(n log n), in-place, unstable
Explanation: Heap sort is in-place but not stable; complexity is O(n log n).

Q87. Which of the following **is immutable in Python**?
A) List
B) Tuple
C) Set
D) Dictionary

Answer: B) Tuple
Explanation: Tuples cannot be modified after creation; lists, sets, dicts are mutable.

Q88. Which OS layer handles **logical addressing**?
A) Physical
B) Data Link
C) Network
D) Application

Answer: C) Network
Explanation: Network layer handles logical addressing (IP addresses).

Q89. Which of the following is **used in BFS traversal of a tree**?
A) Stack
B) Queue
C) Priority Queue
D) Graph

Answer: B) Queue
Explanation: BFS visits nodes level by level using a queue.

Q90. Which of the following **is true about TCP?**
A) Connectionless
B) Unreliable
C) Provides error recovery
D) Stateless

Answer: C) Provides error recovery
Explanation: TCP ensures reliable, ordered data delivery with error checking.

Q91. Which of the following is a **linear search best-case complexity**?
A) O(n)
B) O(1)
C) O(log n)
D) O(n log n)

Answer: B) O(1)
Explanation: Best case occurs when the element is the first in the array.

Q92. Which of the following is **true about adjacency matrix**?
A) Space complexity $O(V^2)$
B) Space complexity O(E)
C) Only used for trees
D) Cannot represent weighted graphs

Answer: A) Space complexity $O(V^2)$
Explanation: Adjacency matrix uses V×V space for V vertices.

Q93. Which of the following **is a non-linear data structure**?
A) Array
B) Stack
C) Queue
D) Graph

Answer: D) Graph
Explanation: Graphs are non-linear, with nodes and edges in any relation.

Q94. Which of the following **is mutable in Java**?
A) String
B) StringBuilder
C) Integer
D) Float

Answer: B) StringBuilder
Explanation: StringBuilder can be modified; String and wrapper classes are immutable.

Q95. Which of the following is **true for a priority queue**?
A) Follows FIFO strictly
B) Always sorted internally
C) Elements are removed based on priority
D) Cannot be implemented with heaps

Answer: C) Elements are removed based on priority
Explanation: Priority queue dequeues element with highest/lowest priority first.

Q96. What is the output of the following Python snippet?

```
a = [1,2,3]

b = a.copy()

b.append(4)

print(a)
```

A) [1,2,3]
B) [1,2,3,4]
C) Error
D) [4]

Answer: A) [1,2,3]
Explanation: .copy() creates a shallow copy; changes to b don't affect a.

Q97. Which of the following **is an example of abstraction**?
A) Using private variables
B) Implementing an interface
C) Inheriting a class
D) Looping over array

Answer: B) Implementing an interface
Explanation: Interface exposes only necessary methods, hiding implementation.

Q98. Which of the following is **true for a circular linked list**?
A) Last node points to null
B) Last node points to first node
C) No head node required
D) Cannot be traversed

Answer: B) Last node points to first node
Explanation: Circular linked list loops back from last node to first node.

Q99. Which of the following **is true about REST API**?
A) Server maintains session
B) Stateless
C) Requires SOAP
D) Connection-oriented

Answer: B) Stateless
Explanation: REST APIs are stateless; each request is independent.

Q100. Which of the following is **true for a max-heap**?
A) Root is smallest
B) Each parent ≥ children
C) Heap is always sorted
D) All nodes have two children

Answer: B) Each parent ≥ children
Explanation: Max-heap property ensures every parent node is ≥ its children.

Q1. What is the time complexity of **binary search** on a sorted array of size n?
A) O(n)
B) O(log n)
C) O(n log n)
D) O(1)

Answer: B) O(log n)
Explanation: Each comparison halves the search space → logarithmic time.

Q2. What is the **worst-case time complexity of Bubble Sort** for n elements?
A) O(n)
B) O(log n)
C) O(n^2)
D) O(n log n)

Answer: C) O(n^2)
Explanation: Two nested loops → n*(n-1)/2 comparisons → O(n^2).

Q3. What is the **space complexity of Merge Sort** for n elements?
A) O(1)
B) O(log n)
C) O(n)
D) O(n log n)

Answer: C) O(n)
Explanation: Requires extra array of size n to merge subarrays.

Q4. What is the **time complexity of accessing an element in an array by index**?
A) O(1)
B) O(n)
C) O(log n)
D) O(n log n)

Answer: A) O(1)
Explanation: Arrays have direct indexing; access is constant time.

Q5. What is the **time complexity of inserting an element at the end of a linked list** (singly linked list, with tail pointer)?
A) O(1)
B) O(n)
C) O(log n)
D) O(n^2)

Answer: A) O(1)
Explanation: With a tail pointer, we can insert at the end in constant time.

Q6. What is the **time complexity of inserting an element at the beginning of a linked list**?
A) O(1)
B) O(n)
C) O(log n)
D) $O(n^2)$

Answer: A) O(1)
Explanation: Only need to update head pointer → constant time.

Q7. What is the **space complexity of Quick Sort (average case)**?
A) O(1)
B) O(log n)
C) O(n)
D) O(n log n)

Answer: B) O(log n)
Explanation: Recursion stack depth is log n in average case; in-place sorting uses no extra array.

Q8. What is the **time complexity of searching an element in a balanced BST**?
A) O(n)
B) O(log n)
C) $O(n^2)$
D) O(1)

Answer: B) O(log n)
Explanation: Each comparison moves to left/right subtree, halving search space.

Q9. What is the **space complexity of recursive Fibonacci** function?
A) O(1)
B) O(n)
C) $O(n^2)$
D) O(log n)

Answer: B) O(n)
Explanation: Recursion stack depth is n for naive recursive Fibonacci.

Q10. What is the **time complexity of traversing a linked list of n elements**?
A) O(1)
B) O(log n)
C) O(n)
D) $O(n^2)$

Answer: C) O(n)
Explanation: Each node must be visited once → linear time.

Q11. What is the **best-case time complexity of Quick Sort**?
A) O(n)
B) O(n log n)
C) O(n^2)
D) O(log n)

Answer: B) O(n log n)
Explanation: Best case: pivot divides array evenly → recursive calls are log n levels.

Q12. What is the **time complexity of inserting at the end of a dynamic array (amortized)**?
A) O(1)
B) O(n)
C) O(log n)
D) O(n^2)

Answer: A) O(1)
Explanation: Most insertions are O(1); occasional resize is O(n), amortized over all insertions → O(1).

Q13. What is the **time complexity of BFS on a graph with V vertices and E edges**?
A) O(V + E)
B) O(V^2)
C) O(E^2)
D) O(V * E)

Answer: A) O(V + E)
Explanation: Each vertex and edge is processed once.

Q14. What is the **time complexity of DFS on a graph with V vertices and E edges**?
A) O(V)
B) O(E)
C) O(V + E)
D) O(V * E)

Answer: C) O(V + E)
Explanation: Similar to BFS, visits all vertices and edges once.

Q15. What is the **space complexity of an adjacency matrix** for a graph with V vertices?
A) O(V)
B) O(E)

C) $O(V^2)$
D) O(V + E)

Answer: C) $O(V^2)$
Explanation: Matrix requires V×V space regardless of edges.

Q16. What is the **space complexity of an adjacency list** for a graph with V vertices and E edges?
A) O(V)
B) O(E)
C) O(V + E)
D) $O(V^2)$

Answer: C) O(V + E)
Explanation: Each vertex stores its neighbors → V + E space.

Q17. What is the **time complexity of inserting into a hash table (average case)**?
A) O(1)
B) O(n)
C) O(log n)
D) $O(n^2)$

Answer: A) O(1)
Explanation: Average case insertion in hash table with good hash function is constant.

Q18. What is the **time complexity of deleting the minimum element from a min-heap**?
A) O(1)
B) O(log n)
C) O(n)
D) O(n log n)

Answer: B) O(log n)
Explanation: Remove root and heapify → log n time.

Q19. What is the **time complexity of searching in a sorted array using linear search**?
A) O(1)
B) O(n)
C) O(log n)
D) $O(n^2)$

Answer: B) O(n)
Explanation: Must check elements sequentially → linear time.

Q20. What is the **time complexity of accessing the i-th element in a singly linked list**?
A) O(1)
B) O(n)
C) O(log n)
D) $O(n^2)$

Answer: B) O(n)
Explanation: Must traverse from head → linear time.

✅ **Key Takeaways for MCQs on Time & Space Complexity:**

1. Arrays → O(1) access, O(n) insertion/deletion
2. Linked List → O(1) insertion at head/tail (if tail pointer exists), O(n) access
3. Stack/Queue → O(1) for push/pop/enqueue/dequeue
4. Sorting → Know complexities (Merge: O(n log n), Bubble/Selection: $O(n^2)$)
5. Recursion → Space complexity = recursion depth
6. Graph → BFS/DFS = O(V + E); adjacency matrix $O(V^2)$, adjacency list O(V + E)
7. Hash Table → O(1) average, O(n) worst-case

8. 1. Sorting Algorithms

Algorithm	Best Case	Average Case	Worst Case	Space Complexity	Stable?	Notes / Tricks
Bubble Sort	O(n)	$O(n^2)$	$O(n^2)$	O(1)	Yes	Compare adjacent and swap; optimized version breaks early if no swaps
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	O(1)	No	Select min each pass; swaps only n times
Insertion Sort	O(n)	$O(n^2)$	$O(n^2)$	O(1)	Yes	Good for nearly sorted arrays
Merge Sort	O(n log n)	O(n log n)	O(n log n)	O(n)	Yes	Divide and merge; not in-place
Quick Sort	O(n log n)	O(n log n)	$O(n^2)$	O(log n)	No	Pivot divides array; worst-case when already sorted (naive pivot)
Heap Sort	O(n log n)	O(n log n)	O(n log n)	O(1)	No	Uses binary heap; in-place
Counting Sort	O(n + k)	O(n + k)	O(n + k)	O(k)	Yes	Only for integers; k = max value
Radix Sort	O(nk)	O(nk)	O(nk)	O(n + k)	Yes	Sort by digits; stable
Bucket Sort	O(n + k)	O(n + k)	$O(n^2)$	O(n + k)	Yes	Uniform distribution assumption

9.

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10. 2. Searching Algorithms

Algorithm	Best Case	Average Case	Worst Case	Space Complexity	Notes / Tricks
Linear Search	O(1)	O(n)	O(n)	O(1)	Check elements sequentially
Binary Search	O(1)	O(log n)	O(log n)	O(1) / O(log n) recursive	Array must be sorted; recursive uses stack O(log n)
Jump Search	O(1)	O(√n)	O(√n)	O(1)	Works on sorted arrays; jump √n steps
Interpolation Search	O(1)	O(log log n)	O(n)	O(1)	Works if values uniformly distributed
Exponential Search	O(1)	O(log n)	O(log n)	O(1)	Uses binary search after finding range
Fibonacci Search	O(1)	O(log n)	O(log n)	O(1)	Like binary search, uses Fibonacci numbers

Common OA Patterns:

- **Arrays / Strings → Sliding Window / Prefix Sum**
- **Sorting → Two pointers / Binary Search**
- **Hashing → Frequency counting, duplicate detection, two-sum**
- **Stack → Next greater element, expression evaluation**
- **Queue → BFS / Level order traversal**
- **Heap → Top K elements, min/max problems**
- **Graph → BFS, DFS, shortest path (Dijkstra, BFS for unweighted)**